

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (1 Question × 100 Marks)
CO1 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
Student Name:	BHARANI DHARAN S	Reg. No.:	192424141
Section / Batch:	192424141	Date:	17/07/2026

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q1	100
GRAND TOTAL:		100 Marks	

SECTION A – Technical Presentation

Q1. Visualization-Based Technical Presentation on Growth Functions

Analyze growth functions $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$, and $O(2^n)$ using visualization tools in a technical presentation format. Clearly define the purpose of comparing these functions for understanding algorithm efficiency. Use graphs, tables, or charts to show how each function behaves as the input size increases. Explain which functions remain practical for large-scale problems and justify your reasoning. Present the conclusions clearly and professionally by linking growth behavior to algorithm selection.

DESIGN AND ANALYSIS OF ALGORITHMS

Foundations of Algorithm Design & Analysis

A Course Overview of Design Paradigms and Analysis Techniques

Presented by

Bharani Dharan S

Reg No: 192424141

Activate Windows
Go to Settings to activate Windows.

01 - INTRODUCTION

What Is DAA, and Why Does It Matter?

1

Defining the field

Design and Analysis of Algorithms (DAA) is the study of constructing efficient algorithms and rigorously evaluating their performance.

2

Correctness comes first

An algorithm must be proven to produce the correct result for every valid input, not just the cases that happen to be tested.

3

Measuring efficiency

Analysis quantifies time and space requirements in a way that is independent of hardware, compiler, or programming language.

4

Classifying problems

DAA groups problems by structure — sorting, searching, graphs, optimization — to match each with a suitable design technique.

5

A foundation for computing

Core DAA concepts underlie compilers, databases, networking, and AI, making them essential across computer science.

Activate Windows
Go to Settings to activate Windows.

How Algorithm Performance Is Analyzed

Concept	What It Measures	Common Notation / Example
Time Complexity	Number of basic operations relative to input size n	$O(n)$, $O(\log n)$, $O(n^2)$
Space Complexity	Additional memory used beyond the input itself	In-place $O(1)$ vs. auxiliary $O(n)$
Asymptotic Notation	Upper, lower, and tight bounds on growth rate	Big-O, Big-Ω, Big-Θ
Case Analysis	Behavior across different categories of input	Best case, average case, worst case
Recurrence Relations	Running time of recursive algorithm designs	$T(n) = 2T(n/2) + O(n)$

Purpose: these techniques give a language-independent, hardware-independent way to compare algorithm efficiency.

Activate Windows
Go to Settings to activate Windows.

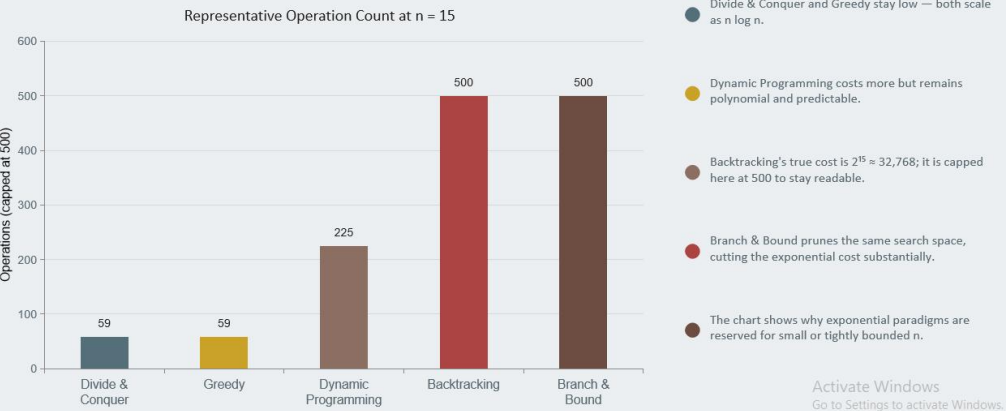
Core Algorithm Design Paradigms

Paradigm	Core Idea	Example Algorithms	Typical Complexity
Divide & Conquer	Break into independent subproblems, solve, then combine	Merge Sort, Quick Sort	$O(n \log n)$
Greedy Method	Make the locally optimal choice at each step	Prim's, Kruskal's, Huffman Coding	$O(n \log n)$
Dynamic Programming	Store subproblem results to avoid recomputation	0/1 Knapsack, LCS, Floyd–Warshall	Polynomial
Backtracking	Build solutions incrementally, abandon invalid paths	N-Queens, Sudoku Solver	Exponential
Branch & Bound	Systematic search pruned using computed bounds	Traveling Salesman, Job Assignment	Exponential (pruned)

Choosing a paradigm depends on problem structure: independence, overlapping subproblems, or the need for exhaustive search.

Activate Windows
Go to Settings to activate Windows.

Complexity Cost of Each Paradigm at Scale



CONCLUSION

Choosing the Right Paradigm and Analysis

- 1 Match the paradigm to problem structure: independent subproblems favor Divide & Conquer; overlapping subproblems favor Dynamic Programming.
- 2 Use Greedy methods only when a problem provably has the greedy-choice and optimal-substructure properties.
- 3 Always analyze worst-case complexity before deployment, especially for Backtracking and Branch & Bound designs.
- 4 Prefer polynomial-time paradigms whenever one exists; reserve exponential approaches for small, bounded instances.
- 5 Effective algorithm design balances correctness, efficiency, and the practical constraints of the target system.

Code of Conduct

I BHARANI DHARAN S (102424141) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____


**RUBRICS FOR CALCULATION**

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1	4	4	4	2	3	90
Overall Total (100)						90

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Activate Windows
Go to Settings to activate Windows.